==
ATTENDX MOBILE - KOTLIN MULTIPLATFORM (KMP) SPECIFICATION & MASTER PROMPT GUIDE
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Overview:
Target Platforms: Android and iOS
UI Framework: Compose Multiplatform (Material 3)
Architecture: Clean Architecture (Presentation, Domain, Data) with MVVM / MVI
Networking: Ktor Client (ContentNegotiation, Kotlinx.Serialization, Bearer Auth)
Local Storage: Multiplatform Settings (Encrypted) + SQLDelight / Room KMP
Dependency Injection: Koin KMP
Location Engine: Google FusedLocation (Android) & CoreLocation (iOS) via expect/actual
Target Backend: Next.js API Routes & Supabase Backend

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1. REST API CONTRACT SPECIFICATIONS

1.1 Authentication & Session
- Endpoint: POST /auth/v1/token?grant_type=password (Supabase Auth / Next.js)
  Headers:
    apikey: <SUPABASE_ANON_KEY>
    Content-Type: application/json
  Request:
    {
      "email": "employee@example.com",
      "password": "secure_password"
    }
  Response (200 OK):
    {
      "access_token": "eyJhbGciOi...",
      "refresh_token": "v-d8fsf...",
      "user": {
        "id": "e4f53c12-32...",
        "email": "employee@example.com"
      }
    }

1.2 User Profile
- Endpoint: GET /api/user/profile
  Headers:
    Authorization: Bearer <access_token>
  Response (200 OK):
    {
      "id": "e4f53c12-32...",
      "name": "Employee Name",
      "email": "employee@example.com",
      "role": "EMPLOYEE",
      "department": "IT Support"
    }

1.3 Geofence Office Locations
- Endpoint: GET /api/locations
  Headers:
    Authorization: Bearer <access_token>
  Response (200 OK):
    [
      {
        "id": "loc_colombo_01",
        "name": "Head Office - Colombo",
        "latitude": 6.9271,

```
      "longitude": 79.8612,
      "radius": 150.0
    }
  ]
```

1.4 Attendance Punch (Check-In & Check-Out)
- Endpoint: GET /api/attendance
  Headers:
    Authorization: Bearer <access_token>
  Response: List of user attendance logs.

- Endpoint: POST /api/attendance
  Headers:
    Authorization: Bearer <access_token>
    Content-Type: application/json
  Request:
```
  {
    "locationId": "loc_colombo_01",
    "latitude": 6.92714,
    "longitude": 79.86121,
    "type": "CHECK_IN"
  }
```
  (Note: type can be either "CHECK_IN" or "CHECK_OUT")
  Response (200 OK):
```
  {
    "success": true,
    "message": "Punch recorded successfully",
    "isLate": false
  }
```

1.5 Gate Pass / Admin QR Punch
- Endpoint: POST /api/admin/scan-punch
  Headers:
    x-gate-passcode: HR
    Content-Type: application/json
  Request:
```
  {
    "userId": "user_guid_123",
    "locationId": "loc_colombo_01"
  }
```
  Response (200 OK):
```
  {
    "success": true,
    "punchType": "CHECK_IN",
    "userName": "Employee Name"
  }
```

1.6 Leave Management
- Endpoint: GET /api/leave-balances
  Response:
```
  [
    { "type": "ANNUAL", "allocated": 14, "used": 3, "remaining": 11 },
    { "type": "CASUAL", "allocated": 7, "used": 2, "remaining": 5 },
    { "type": "MEDICAL", "allocated": 7, "used": 0, "remaining": 7 }
  ]
```

- Endpoint: POST /api/leave-requests
  Request:
```
  {
    "leaveTypeId": "leave_type_guid",
    "startDate": "2026-10-15",
```

```
    "endDate": "2026-10-16",
    "reason": "Family emergency"
  }
```

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2. GOOGLE AI STUDIO MASTER PROMPT (PLAIN TEXT READY)

(Copy the section below directly into Google AI Studio chat or system prompt)

ROLE AND OBJECTIVE:
You are an expert Principal Mobile Software Architect specializing in Kotlin Multiplatform (KMP) and Compose Multiplatform for Android and iOS.
Your goal is to design and develop a production-ready, beautiful, modern Geofencing Attendance & HR Employee Self-Service Mobile Application named 'AttendX'.
The backend is powered by Next.js API Routes and Supabase.

TECH STACK GUIDELINES:
- Target: Android and iOS via Compose Multiplatform (Jetpack Compose shared UI)
- UI Design: Material 3, Dark/Light mode, Smooth animations, Card components
- State Management: MVVM / MVI using ViewModel and StateFlow / SharedFlow
- Networking: Ktor Client (CIO for Android / Darwin for iOS), Kotlinx Serialization
- Storage: Multiplatform Settings (Encrypted for auth tokens) and Room/SQLDelight
- Dependency Injection: Koin Multiplatform
- Location: expect/actual pattern (Google FusedLocation on Android, CLLocation on iOS)
- Navigation: Compose Multiplatform Navigation or Voyager

ARCHITECTURE AND QUALITY STANDARDS:
- Follow strict Clean Architecture:
  1. Data layer: DTOs, API services, Repository implementations, Data sources
  2. Domain layer: Business models, Repository interfaces, Use cases
  3. Presentation layer: UI screens, Components, ViewModels, UI State, UI Events
- Keep common code in commonMain; native bridges strictly in androidMain / iosMain.
- Include proper error handling, network failure recovery, loading states, and offline queueing.

MODULE 1: PROJECT SETUP & FOUNDATION

Tasks to produce:
1. Complete Gradle scripts (root build.gradle.kts and composeApp/build.gradle.kts) with all multiplatform plugins and dependencies configured.
2. Setup Ktor Client singleton with JSON serialization, logging, and token refresh.
3. Setup Koin dependency injection modules in commonMain.
4. Setup generic Result/Resource class and ApiError exception wrapper.

MODULE 2: AUTHENTICATION & SECURE SESSION

Tasks to produce:
1. Data models: AuthRequest, AuthResponse, UserProfile, SessionToken.
2. TokenStorage interface in commonMain with expect/actual implementations:
   - Android: EncryptedSharedPreferences
   - iOS: Keychain via Multiplatform Settings
3. AuthRepository and LoginUseCase handling login, auto-refresh, and logout.
4. Presentation:
   - Modern Splash Screen (session check and auto redirect).
   - Premium Login Screen (animated inputs, biometric prompt hook, error display).
   - AuthViewModel with StateFlow (Idle, Loading, Success, Error).

MODULE 3: LOCATION & GEOFENCING CORE (EXPECT / ACTUAL)

Tasks to produce:
1. Common interface LocationTracker with suspend function getCurrentLocation()
   and observeLocationUpdates() flow.
2. Android implementation in androidMain using Google Play Services
   FusedLocationProviderClient with runtime permission checks.
3. iOS implementation in iosMain using CLLocationManager with delegate callbacks.
4. Geofencing calculation helper in commonMain: Haversine distance formula to
   verify whether coordinates fall within an office radius in meters.

MODULE 4: ATTENDANCE & REAL-TIME CHECK-IN / CHECK-OUT

Tasks to produce:
1. API Service and Repository for fetching locations, attendance history,
   and posting check-in/check-out punch records.
2. Offline Queue: local persistence of pending punch records when offline.
3. Presentation (Dashboard & Punch UI):
   - Circular animated punch button displaying current attendance status.
   - Live Geofence status card with real-time distance and color indicators.
   - Digital clock showing local time and shift details.
   - History list showing today's punch timestamps and total work hours.

MODULE 5: QR SCANNER PUNCH (ADMIN / GATEPASS MODE)

Tasks to produce:
1. Camera/QR scanner Compose component for Android (CameraX) and iOS (AVFoundation).
2. API integration for POST /api/admin/scan-punch using x-gate-passcode header.
3. Scanner UI with animated laser scanner overlay, beep/haptic feedback,
   and employee verification confirmation card.

MODULE 6: LEAVE MANAGEMENT & EMPLOYEE DASHBOARD

Tasks to produce:
1. Models: LeaveBalance, LeaveRequest, LeaveType.
2. API integration for GET /api/leave-balances and POST /api/leave-requests.
3. Presentation:
   - Leave balance visual cards (Annual, Casual, Medical) with progress bars.
   - Leave application sheet with date range picker, leave type selection,
     and reason form.
   - Pending leave requests status list with pull-to-refresh.

EXECUTION INSTRUCTION FOR AI STUDIO:
Begin by producing Module 1 and Module 2 completely with full code, imports, and
directory structure. Wait for confirmation before producing the next modules.
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